Practice Exam - 1

Question 82 of 117

A 39-year-old woman on VA-ECMO for post-cardiotomy shock develops dark, tea-colored urine on hospital day 4. Laboratory studies show lactate dehydrogenase rising from 250 to 1800 U/L, haptoglobin undetectable, plasma free hemoglobin markedly elevated, and a new hyperkalemia of 6.1 mEq/L. There is no fever, and blood cultures from the prior day are negative.

What is the most appropriate next step in management?

A. Start empiric broad-spectrum antibiotics for presumed bloodstream infection

B. Transfuse packed red blood cells and continue current management without further evaluation

C. Inspect the ECMO circuit for thrombus and evaluate for oxygenator/pump exchange

D. Reduce ECMO pump flow alone without further workup

E. Begin therapeutic plasma exchange for presumed thrombotic thrombocytopenic purpura

Commentary | References

Content Code (Blueprint): Critical Care Cardiology — Mechanical Circulatory Support Complications (AHFTC Blueprint)

Task: Interpretation

Teaching Point: Hemolysis is an important non-vascular complication of VA-ECMO, often signaling excessive shear stress from circuit thrombosis, pump malposition, or high negative pressure, and should prompt circuit inspection rather than empiric treatment of an alternative diagnosis.

Correct Answer (Best Answer Choice): Inspect the ECMO circuit for thrombus and evaluate for oxygenator/pump exchange

Distractors (Plausible Incorrect Options):

- Start empiric broad-spectrum antibiotics for presumed bloodstream infection
- Transfuse packed red blood cells and continue current management without further evaluation
- Reduce ECMO pump flow alone without further workup
- Begin therapeutic plasma exchange for presumed thrombotic thrombocytopenic purpura

Rationale:

Correct answer C: (inspect circuit/consider exchange): This presentation (dark urine, rising LDH, undetectable haptoglobin, elevated plasma free hemoglobin, hyperkalemia from cell lysis) is classic for hemolysis, a common complication of ECMO support affecting 5-18% of ECMO supported patients (with higher rates approaching 40% when defined by biochemical alone rather than clinical criteria for hemolysis). The most common causes are thrombus within the oxygenator or pump head, cannula malposition causing high shear, or excessive negative pressure; circuit inspection and exchange of the affected component (if identified) is the appropriate next step.

Incorrect answers:

A. Empiric antibiotics: There is no fever or other evidence of infection, and the laboratory pattern is specific for hemolysis rather than sepsis. While evaluation for infection is reasonable given high infectious rates for VA ECMO supported patients, treating an unconfirmed infection without exploring contributors to hemolysis would not address the cause.
B. Transfusion without further evaluation: Transfusion may be needed to correct anemia caused by ongoing hemolysis. However, transfusion without identifying and correcting the mechanical cause risks ongoing hyperkalemia, renal injury, and circuit failure.
C. Correct answer
D. Reducing flow alone: Lowering flows often reduces shear forces contributing to hemolysis however, it does not identify potential mechanical contributors such as circuit thrombosis that are exacerbating/causing the hemolysis.
E. Plasma exchange for TTP: The clinical and laboratory picture (ECMO setting, normal-presumably normal platelet count as a low platelet count was not described in the stem hyperkalemia from lysis) is consistent with mechanical hemolysis, not a primary thrombotic microangiopathy. Pursuing treatment for TTP could delay appropriate diagnosis of circuit issues and ultimate addressment of contributors to the hemolysis.